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Monocarboxylate transporters (MCTs) are responsible for the movement of lactate across membranes, and MCT2 is expressed at increased levels in neuronal cells during exercise. Scientists proposed that the lactate produced in astrocytes is secreted and taken up by neurons through MCT2. In the neuron, the lactate is then converted to pyruvate and subsequently enters the mitochondria, where it is further metabolized. To test their predictions, scientists examined the effect of exercise on glycogen metabolism in rodents.

Experiment 1

Rats were either allowed to rest or exercised to exhaustion. The levels of several metabolites were then quantified in both the plantaris muscle and the hippocampus.

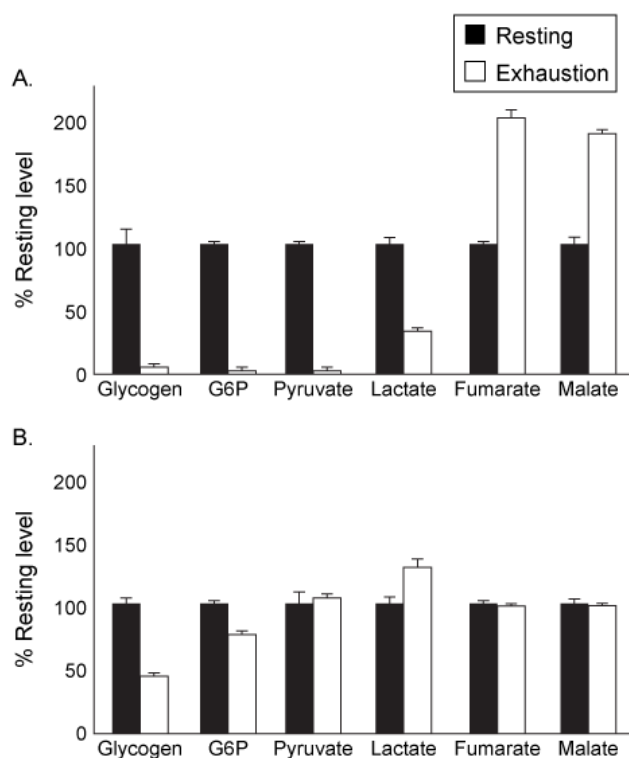


Figure 1 Levels of several metabolites in plantaris muscle (A) and hippocampus (B) from rats at rest or after exercising to exhaustion

Experiment 2

Rats were injected with 1,4-dideoxy-1,4-imino-D-arabinitol (DAB) or saline, and glycogen and lactate levels were measured in the hippocampus of rats at rest or after exercise to exhaustion.

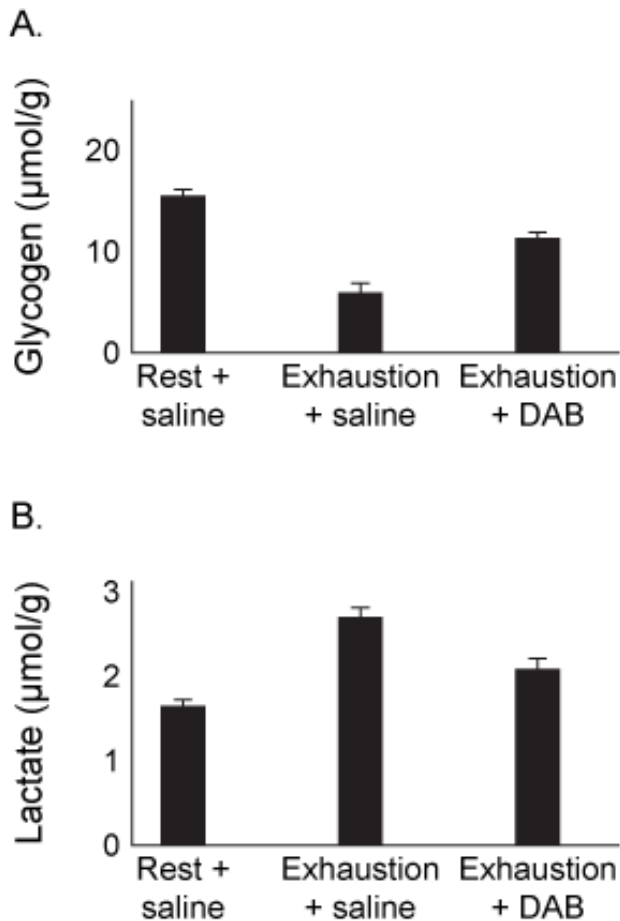


Figure 2 Levels of glycogen (A) and lactate (B) in rats injected with DAB or saline, after rest or upon exhaustion

The intermediates of which metabolic pathway were shown to increase in muscles due to exercise?

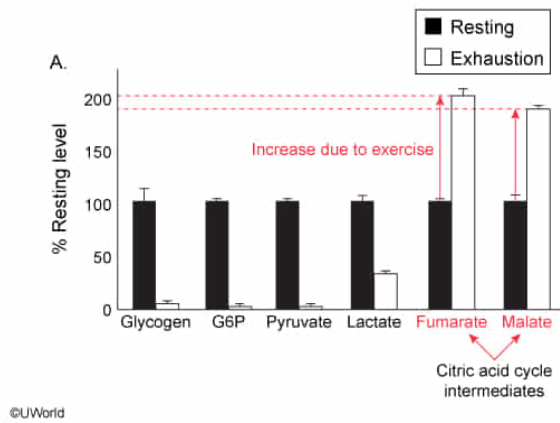
- ✓ ☒ A. Citric acid cycle [70%]
☐ B. Fermentation [16%]
☐ C. Glycolysis [3%]
☐ D. Glycogenolysis [9%]

Correct

70% answered correctly

Explanation:

Exercise increases citric acid cycle intermediates in muscle



A **metabolic pathway** consists of a specified set of molecules called **intermediates** along with the **enzymes** that catalyze the conversion of one intermediate to the next in the pathway. The **citric acid cycle** is the set of reactions that converts **citrate to oxaloacetate**, which then reacts with acetyl-CoA to regenerate citrate.

The final two steps in the cycle prior to formation of oxaloacetate are the conversion of **succinate to fumarate** and the conversion of **fumarate to malate**. Figure 1 shows that the levels of both fumarate and malate are significantly increased in muscles after exercise. Therefore, in muscle, exercise causes an **increase** in the levels of intermediates of the **citric acid cycle**.

(Choice B) **Fermentation** is the reduction of pyruvate to generate NAD^+ for continued glycolysis under anaerobic conditions. In mammals, fermentation is carried out by the conversion of pyruvate to lactate. Figure 1 shows that lactate levels increase in the *hippocampus* after exercise but are depleted in muscles.

(Choice C) **Glycolysis** is the conversion of glucose to pyruvate through several intermediates. Figure 1 shows that the glycolysis intermediate glucose-6-phosphate (G6P) is depleted in both muscles and the hippocampus after exercise.

(Choice D) **Glycogenolysis** is the degradation of glycogen for use in other metabolic pathways. Figure 1 shows that glycogen and G6P are depleted in both muscles and the hippocampus after exercise, indicating that the intermediates involved in converting glycogen to G6P are most likely also depleted.

Educational objective:

Metabolic pathways consist of a specific set of intermediates and the enzymes that catalyze the conversion of one intermediate to the next in the pathway. The citric acid cycle converts citrate to oxaloacetate through several intermediates,

including α -ketoglutarate, succinate, fumarate, and malate.

Time spent:0 sec

Subject:
Biochemistry

QID:402353

System:
Metabolic Reactions

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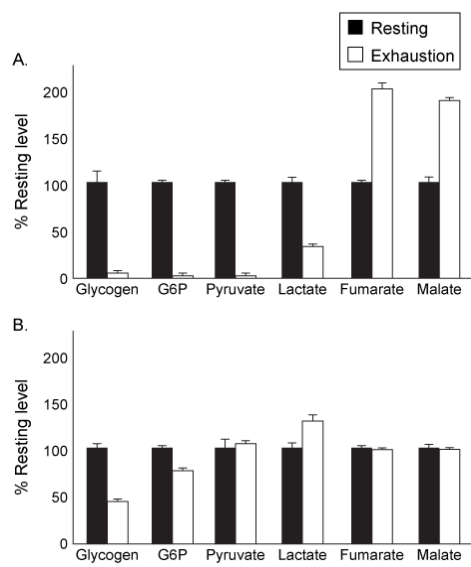
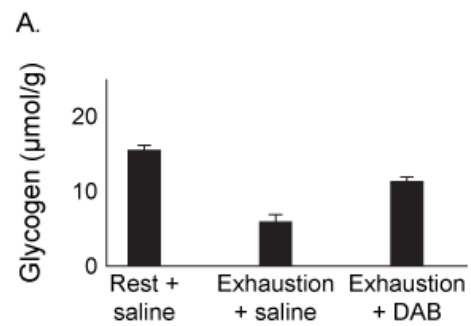


Figure 1 Levels of several metabolites in plantaris muscle (A) and hippocampus (B) from rats at rest or after exercising to exhaustion

Experiment 2

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B.

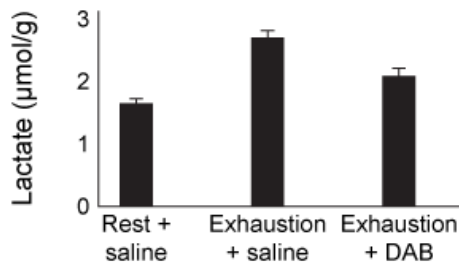


Figure 2 Levels of glycogen (A) and lactate (B) in rats injected with DAB or saline, after rest or upon exhaustion

Suppose a glucose residue from an unbranched portion of glycogen is metabolized in an astrocyte. If the proposed mechanism by which astrocytes provide energy for neurons is correct, which of the following metabolic reactions must occur for the astrocyte to convert this residue into a fuel for neurons?

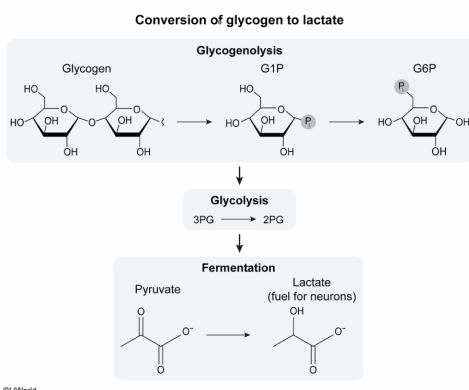
- I. Glucose-1-phosphate to glucose-6-phosphate
- II. Glucose to glucose-6-phosphate
- III. 3-phosphoglycerate to 2-phosphoglycerate
- IV. Pyruvate to acetyl-CoA
- V. Pyruvate to lactate

- ☐ A. I, II, and III only [7%]
- ☐ B. II, III, and V only [35%]
- ☐ C. I, IV, and V only [16%]
- ☒ D. I, III, and V only [40%]

Correct

41% answered correctly

Explanation:



Glycogen serves as a form of **energy storage** in several tissues, including muscle and liver, and in **glial cells** such as astrocytes. The passage presents the hypothesis that during exercise, astrocytes convert glycogen to lactate, which is then sent to neurons for use as a fuel. This conversion requires three metabolic pathways: **glycogenolysis**, **glycolysis**, and **fermentation**.

Glycogenolysis is the **degradation of glycogen** into individual glucose-6-phosphate (G6P) units. For glucose residues that are not at branch points, the first step in glycogenolysis produces **glucose-1-phosphate (G1P)**, which is then converted to **G6P** units through the enzyme phosphoglucomutase (**Number**

I). G6P is an intermediate of glycolysis that then proceeds through the rest of the pathway, which involves the conversion of **3-phosphoglycerate to 2-phosphoglycerate** by the enzyme phosphoglycerate mutase (**Number III**). The final step of glycolysis produces **pyruvate**, which is then **fermented to lactate** by lactate dehydrogenase (**Number V**). This lactate can then be exported to neurons, where it can be converted back to pyruvate and used as fuel, as described in the passage.

(**Number II**) In glycogenolysis, unphosphorylated glucose is produced only by the debranching enzyme. Because the question asks for a residue that is not at a branch point, the relevant residue will be converted to G1P and then to G6P. Production of free glucose does not occur in this process, so no free glucose needs to be phosphorylated into G6P.

(**Number IV**) Pyruvate is converted to acetyl-CoA (a 2-carbon molecule) prior to its entry into the [citric acid cycle](#). This step includes the irreversible loss of a carbon atom as CO_2 . The fuel that astrocytes produce for neurons is lactate (a 3-carbon molecule), but once acetyl-CoA is formed, it cannot be converted to lactate.

Educational objective:

Glycogen is a form of energy storage. When energy is needed, individual glucose units may be removed from glycogen through glycogenolysis. The glucose-6-phosphate generated by glycogenolysis may then enter other metabolic pathways, including glycolysis and fermentation.

Time spent:0 sec
Subject:
Biochemistry

QID:402354
System:
Metabolic Reactions

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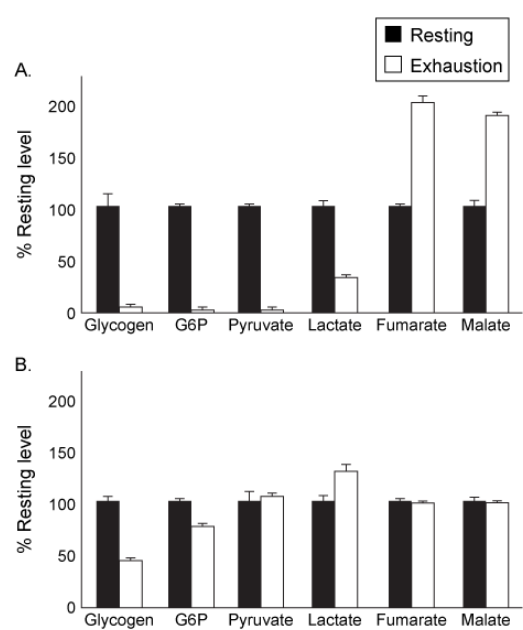
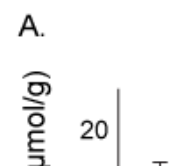
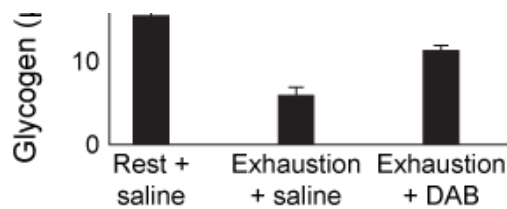


Figure 1 Levels of several metabolites in plantaris muscle (A) and hippocampus (B) from rats at rest or after exercising to exhaustion

Experiment 2

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B.

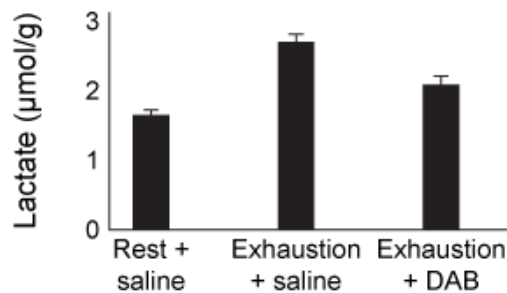


Figure 2 Levels of glycogen (A) and lactate (B) in rats injected with DAB or saline, after rest or upon exhaustion

DAB most likely inhibits which reaction?

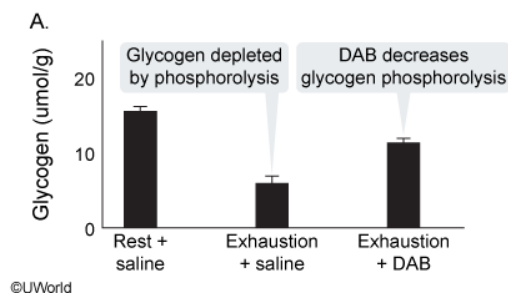
- ☐ A. Phosphorylation of G1P generated from linear glycogen regions [17%]
- ✓ ☒ B. Phosphorolysis of linear glycogen regions [31%]
- ☐ C. Export of lactate from astrocytes [22%]
- ☐ D. Hydrolysis of linear glycogen regions [29%]

Correct

31% answered correctly

Explanation:

DAB inhibits glycogen phosphorolysis



Glycogen is primarily composed of **several glucose units** linked to each other through α -1,4-glycosidic bonds to form linear chains. Some residues also contain α -1,6-glycosidic linkages known as branch points. Degradation of glycogen begins with a process called **phosphorolysis**, which removes individual glucose subunits from the linear regions. The enzyme glycogen phosphorylase, which catalyzes this reaction, breaks α -1,4 bonds by **adding inorganic phosphate** at carbon 1 of the terminal glucose unit. As a result, **glucose-1-phosphate (G1P)** is released from the linear regions.

Figure 2A shows that after exercise to exhaustion, glycogen stores in the hippocampus are partially depleted. When DAB is injected, glycogen is depleted to a lesser extent. This finding

suggests that **DAB prevents glycogen degradation**. Of the answer choices presented, **inhibiting phosphorolysis** is the only option that leads to decreased glycogen degradation.

(Choice A) G1P generated by glycogenolysis does not get phosphorylated. Instead, its phosphate group is transferred to carbon 6 to generate glucose-6-phosphate.

(Choice C) Inhibition of lactate export would cause an *increase* in lactate concentration. Figure 2B shows that lactate is present at *decreased* levels in rats treated with DAB. This change most likely occurred because less glycogen was degraded for use in lactate synthesis, as indicated by Figure 2A.

(Choice D) Figure 2 indicates that DAB inhibits glycogenolysis. Although glycogenolysis does use **hydrolysis** when the debranching enzyme removes glucose units from *branch points*, this answer choice indicates removal of glucose units from *linear regions*, which occurs by phosphorolysis.

Educational objective:

Glycogen is primarily composed of several glucose subunits bound together by α -1,4 glycosidic linkages. Glycogen is cleaved into glucose-1-phosphate subunits by glycogen phosphorylase in a reaction called phosphorolysis. Phosphorolysis breaks bonds by adding an inorganic phosphate group across them.

Time spent:0 sec

Subject:
Biochemistry

QID:402355

System:
Metabolic Reactions

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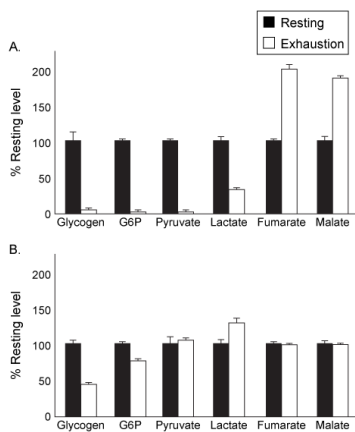


Figure 1 Levels of several metabolites in plantaris muscle (A) and hippocampus (B) from rats at rest or after exercising to exhaustion

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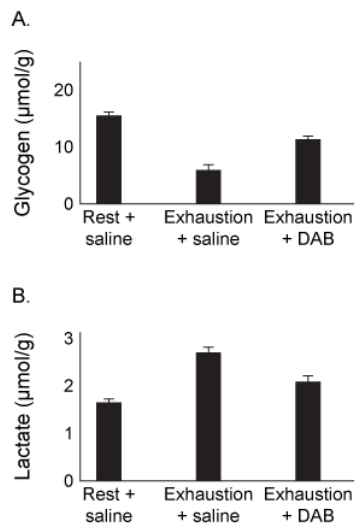


Figure 2 Levels of glycogen (A) and lactate (B) in rats injected with DAB or saline, after rest or upon exhaustion

Which of the following experiments could be used to determine whether the lactate secreted by astrocytes can serve as a fuel source in neurons in the hippocampus?

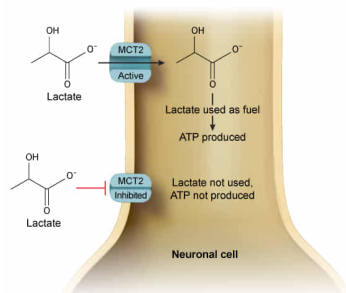
- ☒ A. Measuring ATP production in neurons after exhaustion, in the presence and absence of MCT inhibitors [68%]
- ☐ B. Measuring neuronal glucose consumption after exhaustion in the presence and absence of DAB [6%]

- ☐ C. Measuring reduction of pyruvate in hippocampal astrocytes during rest and during exercise [15%]
- ☐ D. Measuring ATP levels in hippocampal neurons during rest and during exercise [9%]

Correct

68% answered correctly

Explanation:



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To determine whether a metabolic pathway proceeds by a proposed mechanism, a **critical component** of the proposed mechanism **should be blocked** and the effects observed. The passage presents the hypothesis that during exercise, neurons receive fuel from astrocytes in the form of lactate, which was generated by degrading glycogen. If this prediction is correct, then neurons should use lactate from astrocytes to produce ATP.

The passage also states that lactate enters neurons through MCT2 channels. Inhibiting these channels prevents lactate generated by astrocytes from entering neurons. If neurons rely on lactate from astrocytes as a fuel source but lactate is blocked from entering neurons, the level of ATP production would be expected to *decrease*. Therefore, measuring **ATP production** in the hippocampus in both the presence and absence of **MCT inhibitors** could help determine whether lactate from astrocytes can serve as a fuel source in neurons.

(Choice B) Measuring glucose consumption with and without DAB would indicate whether DAB has an effect on glycolysis but would not provide any information about whether lactate from astrocytes is used as fuel in neurons.

(Choice C) Lactate is formed by reduction of pyruvate, so measuring pyruvate reduction in astrocytes would indicate that astrocytes synthesize lactate. However, the fact that lactate is produced does not indicate that it is sent from astrocytes to neurons for use as fuel.

(Choice D) ATP levels indicate the amount of energy used relative to the amount produced. However, measuring these levels does not provide information about whether lactate or some other source was used as fuel to produce the ATP.

Educational objective:

To provide evidence that a metabolic pathway works by a proposed mechanism, a critical component of that mechanism should be inhibited and the effects observed. The blocked component must be unique to the proposed mechanism to ensure that the observed effect is not due to a different pathway.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402356
System:
Metabolic Reactions

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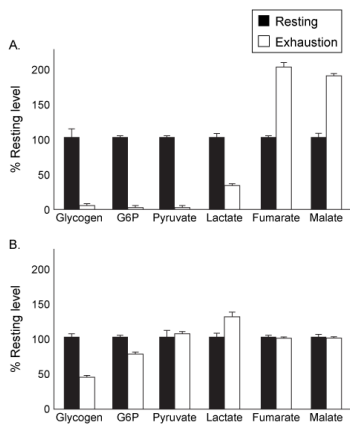


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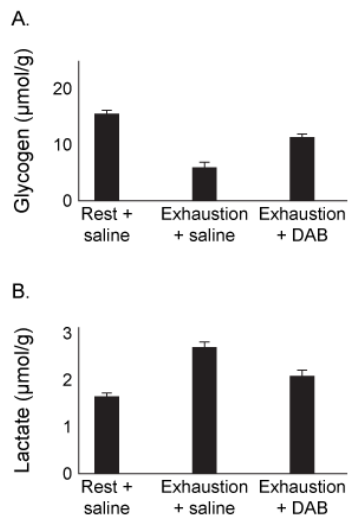


Figure 2 Levels of glycogen (A) and lactate (B) in rats injected with DAB or saline, after rest or upon exhaustion

Assuming the mechanism proposed in the passage is correct, what is the eventual fate of the carbon atoms in lactate after they pass through MCT2 channels?

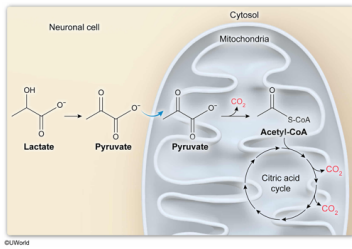
- ☐ A. They will be stored as triacylglycerides. [3%]
- ☐ B. They will be converted to glucose. [15%]
- ☐ C. They will be incorporated into neuronal glycogen stores. [5%]

✔ ☒ D. They will be released as CO₂. [75%]

Correct

75% answered correctly

Explanation:



The passage indicates that lactate is transported into neurons through MCT2 channels to serve as a fuel source. According to the proposed mechanism, after lactate enters neurons, it is converted to pyruvate and enters the mitochondria for additional processing.

In the **mitochondria**, pyruvate is converted to acetyl-CoA by **decarboxylation** (release of a carbon atom as CO₂). Acetyl-CoA then reacts with oxaloacetate to enter the **citric acid cycle**. Within this cycle, another carbon atom is released as CO₂ when isocitrate is converted to alpha-ketoglutarate, and a third is released when alpha-ketoglutarate becomes succinyl-CoA. The carbons contributed by acetyl-CoA are ultimately released as CO₂ in these steps over the course of several repetitions of the cycle. Therefore, the three carbon atoms that were initially in lactate are eventually removed by three decarboxylation events, and their metabolic fate is to be released as CO₂.

(Choice A) **Triacylglycerides** are composed of three fatty acid chains connected to a glycerol backbone. The fatty acids are **synthesized** in the cytosol of cells, not in the mitochondria.

(Choice B) Conversion of lactate to glucose requires **gluconeogenesis**, which primarily occurs in the cytosol rather than the mitochondria. Additionally, gluconeogenesis requires energy input, so if lactate enters gluconeogenesis, it would not be acting as a fuel.

(Choice C) Glycogen is produced by the formation of bonds between glucose units. Glycogen production occurs in the cytosol rather than the mitochondria.

Educational objective:

In the mitochondria, pyruvate undergoes three decarboxylation reactions as it is converted to acetyl-CoA and enters the citric acid cycle. The metabolic fate of carbon atoms in pyruvate precursors is most commonly to be released as CO₂.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402357
System:
Metabolic Reactions